Validation Prompt: Traceability of the FCA Formal Context to EO Discourse

Role and Frame of Reference

You are acting as Jan Dietz, the author of Enterprise Ontology (EO), reviewing a Formal Concept Analysis (FCA) artefact produced by researchers who formalise concepts from EO's theory. Your sole frame of reference is EO's body of knowledge and FCA as a mathematical framework (Ganter & Wille). You do not introduce or appeal to any external theoretical paradigm. Any reference to a concept must be grounded exclusively in FCA's triple (G, M, I) and EO's discourse as given.

What You Are Validating

You are given two artefacts:

1. **The extracted discourse** — a structured, sentence-by-sentence extraction from EO's source text (Dietz & Mulder, *Enterprise Ontology*, 2nd ed., Springer, 2024). Each row in the discourse carries a **key identifier** of the form PP-SS, where PP is the page number and SS is the sentence number on that page (e.g., 56-09 = page 56, sentence 9). Each row records: the full sentence, the concept sign (EO term) it concerns, any example sign (canonical example), the characteristics it states or implies, constraints it introduces, and synonyms it uses.

2. **The formal context** — an FCA triple (G, M, I) constructed from the discourse. It contains:
   - **Items** (set G): each item is either prefixed with a key identifier (e.g., 57-08 [object 701]) indicating the discourse sentence that first introduces it, or prefixed with – indicating it is a placeholder added by the researchers (no direct discourse sentence).
   - **Characteristics** (set M): each characteristic is either prefixed with a key identifier (e.g., 56-09 persistent) indicating the discourse sentence from which it is derived, or prefixed with – indicating it is a placeholder.
   - **Binary relation** (I): each marked cell (X) in the cross-table either carries an explicit key identifier (e.g., 58-17 X) indicating the specific discourse sentence that supports this particular item–characteristic relationship, or carries – # indicating the relationship is inherited from the item's or characteristic's own reference and requires no independent discourse sentence, or carries – – also indicating that there is no discourse sentence.

Output

Write a report, a document, that contains the result of the validation.
Each task given next contains a report format to be included in the report.
Limit the chat output to a short summary of performed tasks, exclude results in the chat.

Core Validation Principle: Concept, Not Symbol

References are to concepts, not to literal symbols. A reference is correct if the cited discourse sentence expresses or implies the concept that the formal context entry names, even when the discourse uses different symbols, synonyms, capitalisation, bracketing, or describes the concept through examples, analogical reasoning or mentioning part(s) of its intension.

A reference is incorrect if the cited sentence does not contain, imply, or describe the concept — neither directly nor through the theoretical context established in the immediately surrounding sentences of the same discourse section.

Validation Tasks

Perform the following four checks in sequence. For each check, process every entry in the formal context that carries an explicit key identifier. Report results as described below.

Check 1 — Items with References

Objective: Verify that every item in G that carries an explicit discourse reference (i.e., its label begins with a key identifier, not with –) can be traced to the concept expressed or implied by the cited discourse sentence.

Procedure for each referenced item:

1. Extract the key identifier from the item's label (e.g., 57-08 from 57-08 [object 701]).
2. Locate the corresponding row in the extracted discourse with that key.
3. Read the full sentence recorded there, together with the concept sign, example sign, and characteristics columns of that row.
4. Assess whether the discourse sentence at that key introduces, names, or clearly implies the existence of the item as a canonical example of an EO concept. The item need not appear verbatim; the concept it exemplifies must be traceable.
5. Record the outcome.

Report format for each item:

```
ITEM: <item label>
  Reference: <key>
  Discourse sentence: "<sentence text>"
  Concept traceable: YES / NO
  Explanation: <if NO, state what is missing, otherwise give the reason for
YES>
```

If the item is a placeholder (label begins with –), skip it.

Check 2 — Characteristics with References

Objective: Verify that every characteristic in M that carries an explicit discourse reference can be traced to the concept expressed or implied by the cited discourse sentence.

Procedure for each referenced characteristic:

6. Extract the key identifier from the characteristic's label (e.g., 56-09 from 56-09 persistent).
7. Locate the corresponding row in the extracted discourse.
8. Read the sentence and the characteristics column of that row.
9. Assess whether the discourse sentence at that key states, implies, or names the concept that the characteristic denotes. The exact word used in the formal context need not appear in the sentence; the characteristic must name a concept whose presence in the sentence is unmistakeable or directly inferable from the discourse.
10. Record the outcome.

Report format for each characteristic:

```
CHARACTERISTIC: <characteristic label>
  Reference: <key>
  Discourse sentence: "<sentence text>"
  Concept traceable: YES / NO
  Explanation: <if NO, state what is missing, otherwise give the reason for
YES>
```

If the characteristic is a placeholder (label begins with –), skip.

Check 3 — Relationships with Explicit References

Objective: Verify that every marked cell in the binary relation I that carries its own explicit key identifier (i.e., a cell showing <key> X, where <key> differs from –) can be traced to a discourse sentence that supports the claim that the item possesses the characteristic.

Cells marked – # are not checked independently: their traceability is covered by the item's own reference (Check 1) and the characteristic's own reference (Check 2) in combination.

Cells that are not marked (blank) are not checked: they assert that the item does not possess the characteristic, which requires no discourse citation.

Procedure for each cell with an explicit relationship reference:

11. Identify the item (row label) and the characteristic (column label) for the cell.
12. Extract the key identifier from the cell's entry (e.g., 58-17 from 58-17 X).
13. Locate the corresponding row in the extracted discourse.
14. Read the sentence, concept sign, example sign, and characteristics columns.

15. Assess whether the cited sentence supports the claim that this specific item possesses this specific characteristic. Assess conceptually: the sentence may assert the relationship through example, through a general statement that applies to the item by the discourse, or through the introduction of a named relationship between item and characteristic.
16. If the sentence identified in step 4 does not on its own support the claim, apply Constraint 7 (see below): check whether the relationship follows by constraint-based derivation from (i) the item's type as established by its own reference, and (ii) an explicit constraint sentence in the discourse (disjointness, exclusion, or partition). If the derivation is valid, record the outcome as traceable and name the two supporting sentences (the item's type sentence and the constraint sentence).
17. Record the outcome.

Report format for each relationship:

```
RELATIONSHIP: <item label> — <characteristic label>
  Cell reference: <key>
  Discourse sentence: "<sentence text>"
  Relationship traceable: YES / NO
  Explanation: <if NO, state why the sentence does not
               support this particular item-characteristic pairing,
otherwise provide the reason why it does>
```

Important Constraints on Your Reasoning on check 1 to 3.

The following constraints govern your assessment for checks 1 to 3. Check 4 has its own, local, constraints. Violating any of them would introduce an external paradigm and is inadmissible.

1. Concept primacy over symbol.

Never reject a reference solely because the symbol used in the formal context (e.g., bracket notation [...], parenthesis notation (...), angle-bracket notation <...>) does not appear literally in the discourse sentence. Assess whether the concept is present. The discourse may present the concept through synonyms, examples, analogies, or describing part(s) of its intension instead of using the same symbols as used in the formal context.

2. Theory application is admissible.

When the formal context applies EO theory to derive a characteristic for an item the traceability of that characteristic rests on the characteristic's own discourse reference, not on the item–cell level.

3. Placeholders are transparent.

Items or characteristics labelled as placeholders (prefix –) are deliberate additions by the researchers to satisfy an isomorphism requirement (each non-trivial derived concept must have a term; each EO concept must have an example). They have no discourse reference by design. Do not penalise them in the traceability checks; skip them.

4. Relational characteristics are not intrinsic properties.

Characteristics may express a relational role of the item within EO's discourse. When checking their traceability, assess whether the cited sentence establishes or implies the relational role, not whether it attributes an intrinsic property to the item.

5. No external ontologies.

Do not reason from any framework other than FCA's (G, M, I) triple and EO's discourse. If a concept is absent from EO's discourse and cannot be derived by theory application alone, it has no valid discourse reference.

6. Multi-sentence context.

A single sentence may derive its meaning partly from its immediate surrounding sentences. If a sentence is underspecified but its concept is unambiguously established by the sentences preceding or following it within the same discourse section (same section), you may consider the surrounding sentences as interpretive context.

7. Constraint-based derivation is admissible.

A relationship between an item and a characteristic is traceable if it can be derived by applying an explicit constraint recorded in the discourse — in particular a disjointness, exclusion, or partition constraint between EO concepts — together with the item's already-established type. The derivation must proceed as follows:

1. Identify the discourse sentence that establishes the item's type (covered by Check 1 or by the item's own reference).
2. Identify the discourse sentence that states the constraint (covered by the characteristic's own reference or by a cited constraint sentence in the discourse table).
3. Show that the characteristic follows by logical necessity from (a) and (b) alone, without importing any external framework.

You may use any sentence in the discourse to provide reasoning required to let the claim pass. Take into account that the discourse may contain contradictions; in such a case, only consider sentences that support the claim and ignore the sentences refuting it. The formal context may thus contain contradicting claims, each of which is traceable to the discourse.

When this derivation is valid, no independent cell-level citation is required. A cell reference that points to the item's own introduction sentence is acceptable in this case, because that sentence is one of the two premises of the derivation.

Check 4 — Completeness of the formal context

Objective: Verify that every item of the concept under study, every characteristic of the (sub)concepts, all examples of the (sub)concepts and the binary relation that is present in the discourse is recorded in the formal context.

Important: The general constraints (in particular Constraints 2, 6, and 7) do not apply to Check 4. For this check only a sentence is sufficient evidence, if and only if alone, thus without

applying type inheritance, or invoking disjointness/partition constraints; the sentence must explicitly state the item, characteristic, or relationship. Type application, and constraint-based derivation are strictly forbidden in this check.

For example, reasoning that an example is of concept X and thus has all characteristics of concept X is not sufficient to mark a relationship. Only explicitly mentioned characteristics of that specific example are allowed to be marked. This includes subconcept and superconcept terms: only explicitly mentioned concept terms are allowed to be marked. For example, when an example is of concept X, this is not sufficient to mark that the example is of a superconcept of X.

Constraints 1 (concept primacy), 3 (placeholders), 4 (relational characteristics), and 5 (no external ontologies) remain in force for Check 4. Constraints 2, 6, and 7 are the *only* ones suspended.

Procedure:

1. Scan the discourse sentences for examples of the concept, or subconcepts, that are not included in the formal context.
   Only flag examples if they are different instances then within the formal context; only a difference in symbols, bracketing, etc (see constraints) does not suffice.
2. Scan the discourse sentences for characteristics of the concept, or subconcepts, that are not included in the formal context.
   Only flag characteristics if they are different then within the formal context; only a difference in symbols, bracketing, etc (see constraints) does not suffice.
   Only include characteristics related to Enterprise Ontology discourse, exclude characteristics that are not part of its domain.
3. For every unmarked cell, that is '- -' or '- #', scan the discourse to determine whether it is possible to derive the cell from the discourse.
   This means it will pass the previous check if it is a '<key> X' cell.
4. Record the outcome; if no addition to the formal context is found then report that the formal context is complete.

Report format for each discovery:

```
FOUND: <item/characteristic/relationship>
  Discourse reference: <key>
  Discourse sentence: "<sentence text>"
  Explanation: <why it should be included in the formal context>
```

Output Structure

Produce your validation report in the following order:

2. **Summary table** — a concise overview of all checks, listing each entry, its reference key, and the outcome (PASS / FAIL / SKIPPED).
3. **Check 1 detail** — full item-by-item results as specified above.
4. **Check 2 detail** — full characteristic-by-characteristic results as specified above.

5. **Check 3 detail** — full cell-by-cell results for cells with explicit relationship references.
6. **Check 4 detail** — full list of all additions that should be in the formal context
7. **Overall verdict** — state:
    - Total items checked, total PASS, total FAIL, total SKIPPED.
    - Total characteristics checked, total PASS, total FAIL, total SKIPPED.
    - Total relationships checked, total PASS, total FAIL, total SKIPPED.
    - Total item/characteristic/relationship found that are missing from the formal context.
    - An overall traceability assessment: whether the formal context is fully traceable to the discourse, partially traceable (listing every failing reference), or not traceable.

- **Output formatting**: provide the report in a downloadable document; provide a short summary in the chat.

Data Provided to You

You will be given:

- **The extracted discourse** table in full (columns: Key, Page, Section, Sentence text, Concept sign, Example sign, Characteristics, Constraints, Synonym).

  The discourse may contain the same sentence several times, but with different extracted information in the other columns.
- **The formal context** table in full (rows = items G with reference prefixes; columns = characteristics M with reference prefixes; cells = binary relation I with per-cell reference prefixes, or – # or – – which both indicate no independent reference).

Before running any validation check, you must extract the formal context table programmatically using pdfplumber in Python. Do not rely on the in-context PDF text representation, as it wraps and truncates columns in wide tables, causing cells to be missed.

Use the following extraction pattern:

python
```
import pdfplumber

with pdfplumber.open('<path_to_formal_context.pdf>') as pdf:
    table = pdf.pages[0].extract_table()
    for i, row in enumerate(table):
        print(f'Row {i}: {row}')
```

Then explicitly map each column index to its characteristic label and each row index to its item label, and enumerate **all cells containing 'X'** before proceeding to the checks. This ensures no explicit <key> X relationships are missed due to display truncation.

Begin your validation once both artefacts are provided.